SMART INDIA HACKATHON 2025

JOCKY

- **Problem Statement ID - 26148**
- **Problem Statement Title -**
  Smart Tourist Safety Monitoring & Incident Response System using AI, Geo-Fencing, and Blockchain-based Digital ID

- **Theme - Travel & Tourism**
- **PS Category- Software**
- **Team ID-**
- **Team Name - KitKat**

Tour Sarthi

- **Proposed Solution (Describe your Idea/Solution/Prototype)**

- Detailed explanation of the proposed solution
- How it addresses the problem
- Innovation and uniqueness of the solution

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Tour Sarthi

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To create a **block-chain** based Tourist Safety System, that ensures **Secure Identity Generation**, a **AI 24x7 Safety Support System** while focusing on **Data Privacy and Security.**

Innovation and Uniqueness

Simulated Decentralized Block-Chain ID
that ensures tamper-proof identity and travel records.

Agentic AI Workflow
Scrapes and analyzes real-time data to identify and score potential safety threats before they impact tourists.

Crowdsourced Verification
Leverages a network of nearby tourists to confirm incidents in real-time, drastically reducing false alarms and ensuring an efficient response.

Security First-by-Design
Maximizes safety with minimal data collection by performing anomaly detection and tracking locally on the user's device.

Real-time Safety Dashboard
Provides authorities with an interactive map showing incident heat maps and tourist clusters for data-driven situational awareness.

Instant SOS Alerting
One-touch panic button instantly transmits live location, camera/mic feeds, and a critical alert to the nearest police unit.

Problems Addressed

- Unique Block Chain based IDs for Tourists. Making Immutable records while maintaining Security.
- Providing AI-based Anomaly Detection, to reduce Crime against Tourists by providing a robust support system
- Data Visualization of Tourists for the Police and Authorities to track and monitor, while not explicitly disclosing private information of tourists.
- Unique Data Security Implementation, that protects the privacy of the users while providing Real-Time data.

Proposed Solution

Uniqueness

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NaVira - Smart safety that travels with you

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Brief of Idea -

A smart **tourist safety monitoring and incident response system**, based on blockchain based identification, multi-model agentic AI system utilizing real-time data, geofencing and a security first architecture along with flexible IoT to ensure 24x7 safety of tourists.

Proposed Solution

NaVira - An all in one tourist safety platform that fits to every environment, a reliable step to integrate modern day tech like AI, blockchain and IoT in current tourism scope for enhanced safety and security.

- **Blockchain** based KYC for **tamper-proof** identity (DID) and travel records
- **Agentic AI workflow** to analyze real-time data to identify potential safety threats
- **Safety Scoring Model** Continuously evaluates tourist risk in real-time using AI
- **Anomaly Detection Model** detects unusual tourist behavior like route deviations, inactivity
- **Security-First Architecture** which Prioritizes privacy and data security at core
- **Digital Laws Compliant,** all data handling meets Indian privacy and data protection regulations.
- The app and alerts provide **multilingual support** for both Indian and global tourist
- Works seamlessly in low-network or remote zones using **offline buffering and IoT integration**.
- Gives **authorities** real-time heatmaps, alerts, and verified safety data for **rapid response**.
- **Integration with Existing solutions** using tourist ID across various platforms

What makes Navira Unique?

Personalized Safety
Panic button instantly transmits live location, camera & mic feeds, thus a critical alert to the nearest police unit.

Continuous Alerts & Notifications
Pushes live updates such as rain or heat alerts, sudden closures of tourist spots, or evolving safety risks in the area.

Crowdsourced and CCTV based Verification
Leverages a network of nearby tourists to confirm incidents in real-time, drastically reducing false alarms and ensuring an efficient response.

Real-time Safety Dashboard
Provides authorities with an interactive map showing incident heat maps and tourist clusters for data-driven situational awareness.

Instant SOS Alerting
One-touch panic button instantly transmits live location, camera/mic feeds, and a critical alert to the nearest police unit.

Problems Addressed

- Tourists face safety concerns like harassment and accidents along with lack of appropriate responses damaging India’s reputation.
- There is no real-time safety monitoring or integrated emergency support for tourists.
- Incident reporting and response are slow, fragmented, and prone to false alarms.
- Tourist data is scattered across platforms with no unified system for safety tracking.
- Remote and low-connectivity zones leave tourists vulnerable without reliable support.

TECHNICAL APPROACH

- Technologies to be used (e.g. programming languages, frameworks, hardware)
- Methodology and process for implementation (Flow Charts/Images/ working prototype)

TECHNICAL APPROACH

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Tourist Mobile App (React Native)

User Layer

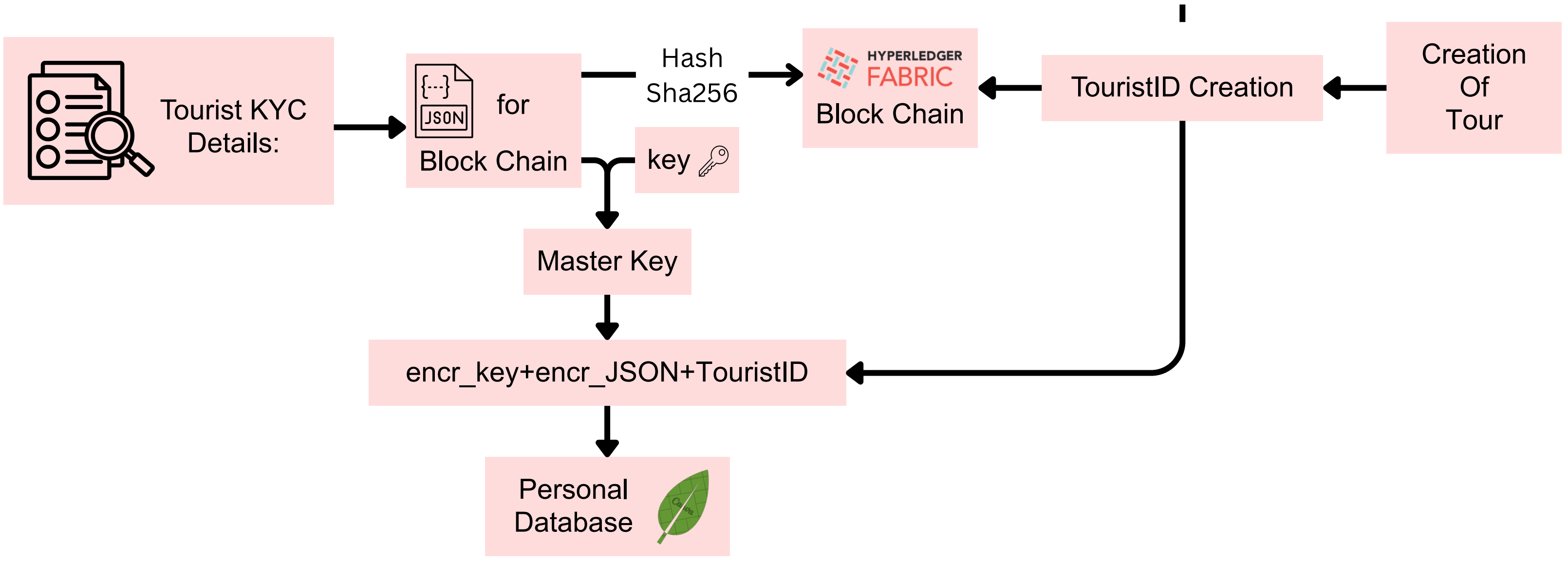
- Real-time Data
- Itinerary Details
- IoT Sensor Data
- Personal Details

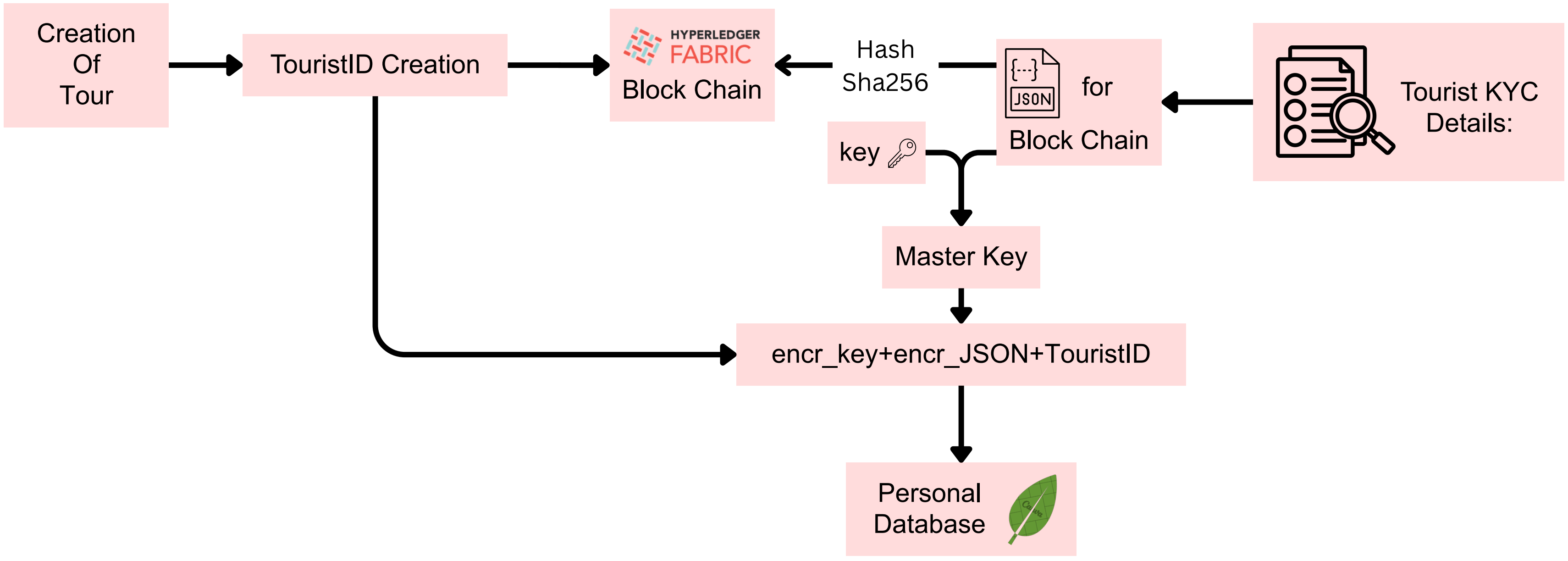
Tourist ID

- Used on booking platforms
- Itinerary Auto Sync

Integration

- Tourist gets a blockchain-based Tourist ID, used for bookings on various platforms.
- A user books a flight, hotel, or tour package using Tourist ID, the details are automatically linked to their profile and itinerary is synced into the App.





TECHNICAL APPROACH

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Blockchain Layer (Hyperledger fabric)

HYPERLEDGER FABRIC

Permissioned blockchain

Blockchain based KYC system

- Used Hyperledger Fabric for permissioned private blockchain
- Automated ID Lifecycle Management via Smart Contracts (Chaincode)
- Granular Access Control via Membership Service Provider (MSP)
- Data Isolation and Scalability via a Multi-Channel Architecture
- Real-time Off-Chain State Synchronization via Block Event Listeners

Tourist ID → Use on booking platforms (flight, trains, hotels) → Itinerary auto sync with our database

Integration with existing solution

- Tourist gets a blockchain-based DID (tourist ID) which can be used for all bookings like flight, hotel etc, on various platforms
- When user books a flight, hotel, or tour package using Tourist ID, the booking details are automatically linked to their profile and itinerary is synced into the App.
- Enables proactive anomaly detection (e.g., "Tourist did not check into hotel at expected time").

User Layer

KYC

- Tourist Mobile App (React Native)
- Authorities/Police dashboard (React / D3.js)
- IoT solutions [LoRa and radio] (for low network zones)

Nginx API gateway

Tourist App

Authorities dashboard

Panic Button : Instantly notifies authorities and nearby tourists when a tourist triggers a panic button or voice-activated alert.

Backend (APIs and services)

Database layer

Writes → Encryption Layer

Database Interface

Secure KMS

Read → Decryption Layer

Main database

- Main database stores tourist profiles, itineraries, and booking details securely including historical incidents and safety records.
- Added **encrypted data storage** for enhanced privacy, using secure **KMS** (key management service)
- Fully **DPDP Act compliant**: minimal data storage, encrypted at rest, and privacy-focused.

AI Engine

Anomalies Detection

Real-time safety scoring

Kafka (Event Bus)

Scraper AI agent (Runs in regular interval for sync with real-time data)

Data sources (Govt APIs, News outlets) → Newspaper / Article → Headline analysis → LLM Parsing → JSON Schema → Geo-tagging using location info (LLM + Google API)

Add data to incident database

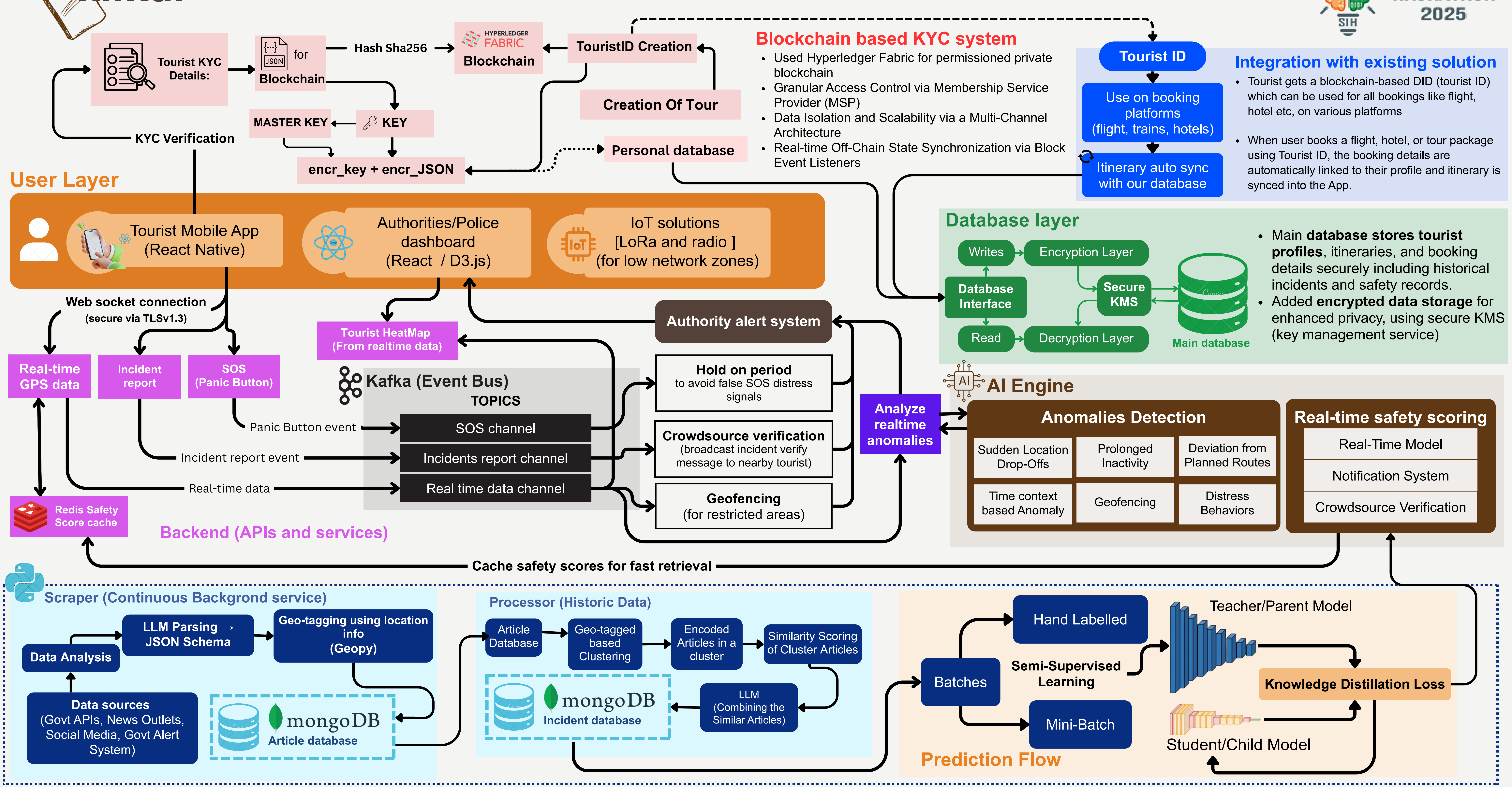
- Collects data about a particular location from public websites, APIs, and social media.
- Analyzes data using **LLM** to identify risk levels and perform **geo-tagging**.
- Feeds insights to the Safety Score Engine and authorities for real-time alerts.

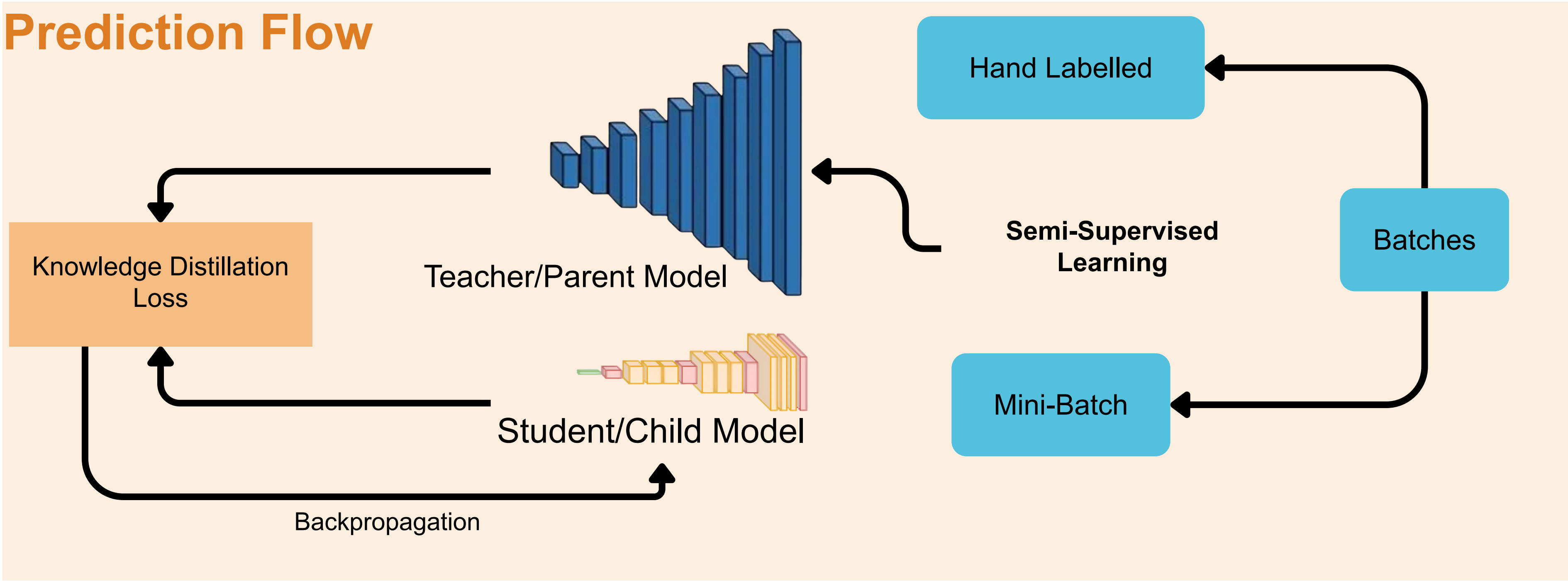
mongoDB

Incidents database (scraped from news and govt APIs)

Feeds context to real-time safety scoring model

TECHNICAL APPROACH



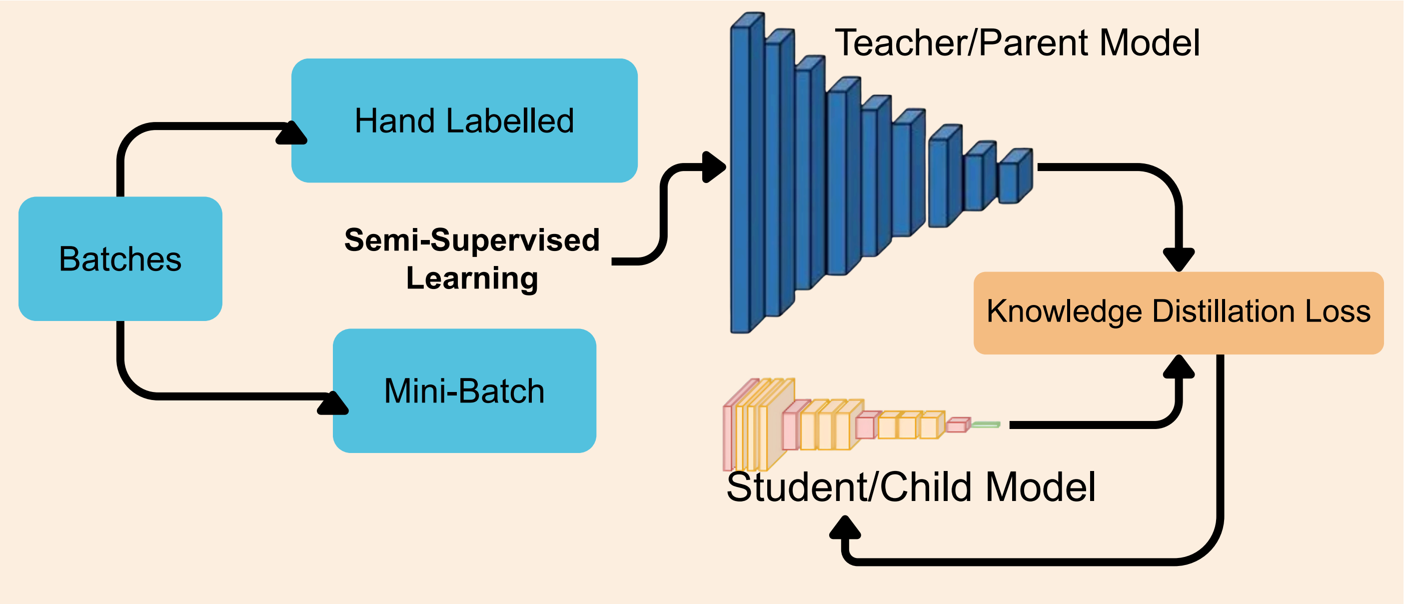


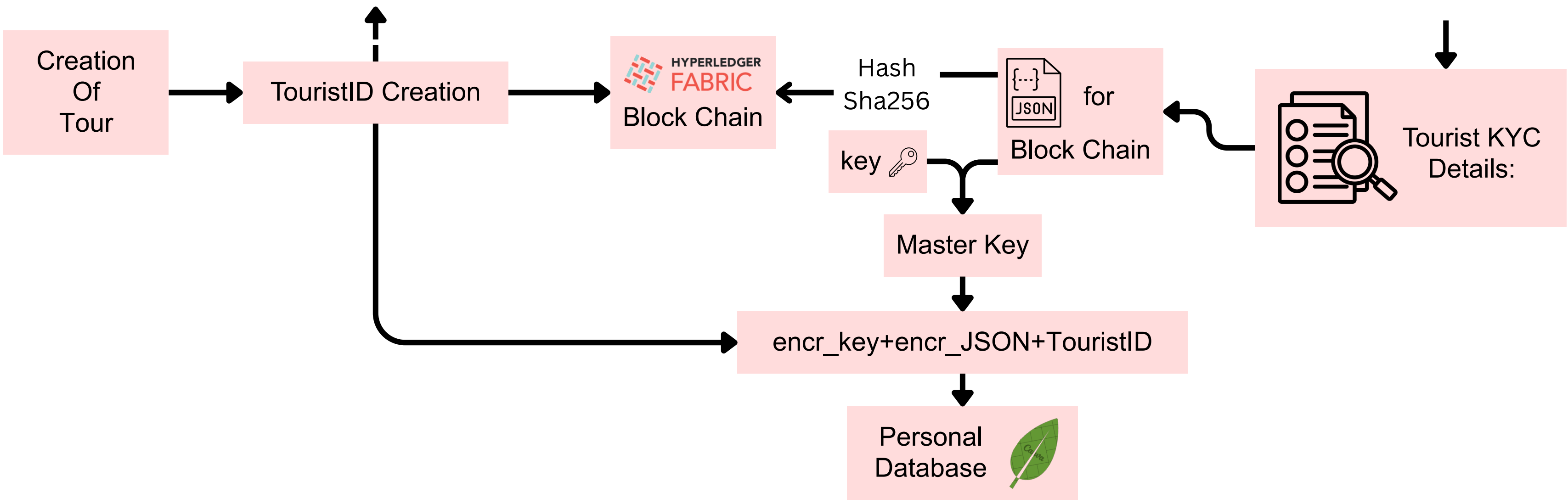
Panic Button
Instantly notifies authorities and nearby tourists when a tourist triggers a panic button or voice-activated alert.

Incident Reporting System
Enables tourists and locals to instantly report threats or incidents for verified, rapid authority response.

Itinerary Planner
Automatically syncs tourist plans with real-time safety scores to suggest safer activities and timings.

Safe Routes
Provides dynamically updated navigation that avoids unsafe zones based on live incident and risk data.





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TECHNICAL APPROACH

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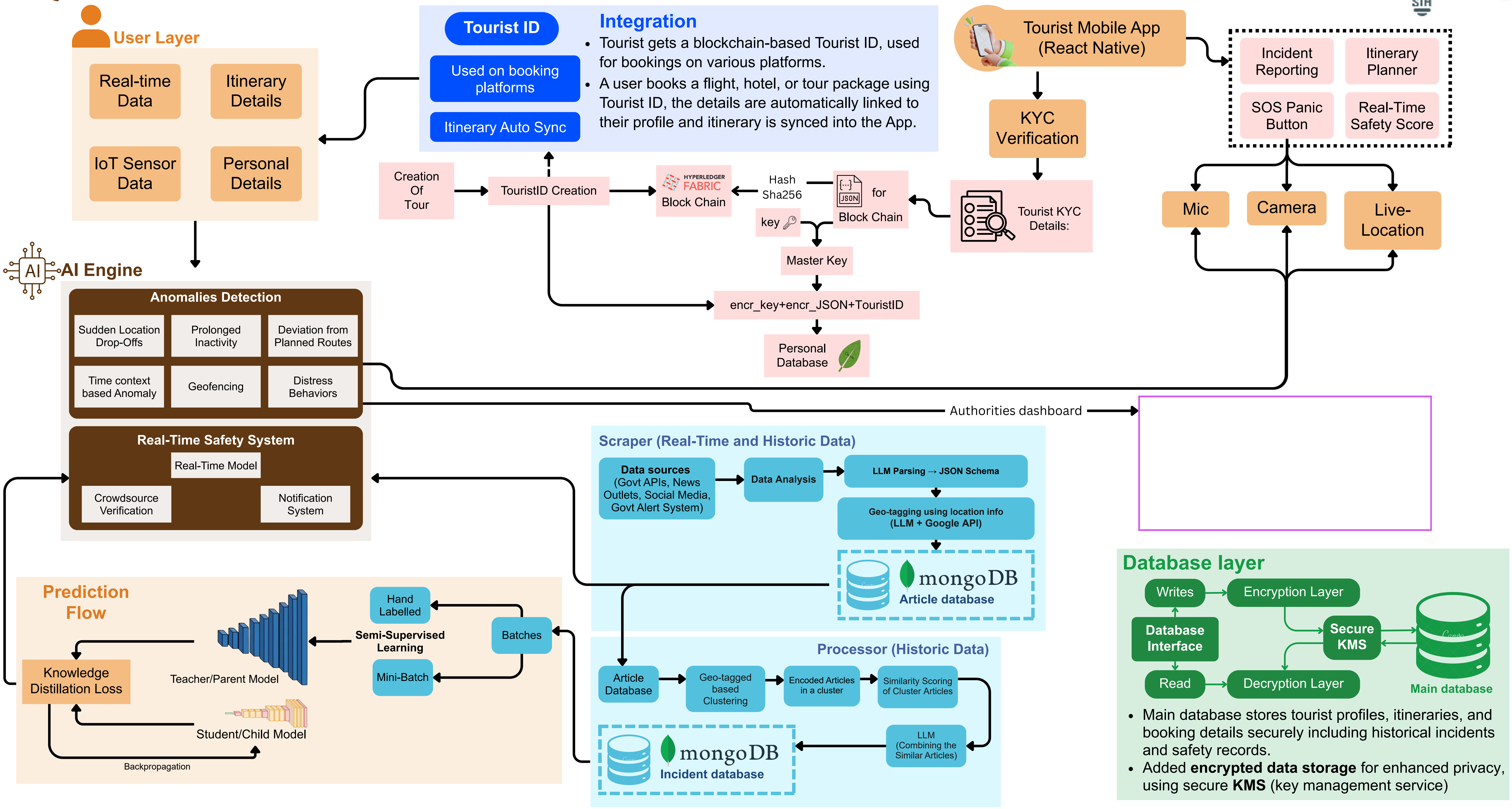

TECHNICAL APPROACH

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Backend (APIs and services)

AI Engine

Geo fencing service

Scraper AI agent

Data sources
- govt API
- news outlets

Incident response service

Notification services

Database layer

SOS (Panic button) Service

FEASIBILITY AND VIABILITY

- Analysis of the feasibility of the idea
- Potential challenges and risks
- Strategies for overcoming these challenges

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FEASIBILITY AND VIABILITY

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Technical Feasibility

The idea is **practical and achievable**, as it uses existing and proven technologies (GPS, AI anomaly detection, blockchain identity, real-time communication) combined in a **novel way for tourist safety.**

Operational Feasibility

Deployable via mobile app for tourists (**multilingual**, **offline support**) and dashboards for authorities (**heatmaps, incident management**). Built on familiar web and app standards.

Infrastrucutural Feasibility

Requires cloud-based backend for **AI models** and event streaming, blockchain for secure ID, and mobile/web apps. **Economically feasible** due to open-source frameworks and pay-as-you-go cloud models that keep costs low while **supporting scalability.**

Large Scale Adoption

The **modular and cloud-native design** supports scaling from city-level pilots to nationwide. Its **interoperability with existing platforms** makes it easy for governments, travel agencies, and hotels to adopt without disruption, fostering trust and global adoption.

Viability

Applicability

Provides actionable real-time safety scores, alerts, and verified incident reports. Supports both urban and remote areas through offline GPS buffering and IoT/radio-frequency integration.

Real-Life Implementation

Adoptable by tourism boards, travel agencies, hotels, and transport providers to monitor tourist safety without altering existing workflows. Tourist ID enables automatic itinerary sync.

Scalability

Designed as a modular, cloud-native system that can start with small pilots (city-level) and scale up to state, national, or international deployments.

Future Scope

Advanced AI/ML techniques can extend the system to predictive safety, such as detecting early signs of crowd stampedes, cloudbursts, or other disasters — enabling preventive alerts before incidents happen.

Challenges & Risks	Our Strategy to Overcome
Data Privacy & Compliance – Handling sensitive tourist data while ensuring DPDP Act compliance.	Adopt a **privacy-first design** with minimal data storage, full encryption, and blockchain-backed critical event logging.
Connectivity in Remote Areas – Real-time monitoring may fail in low-network or no-network zones.	Use **offline GPS buffering and IoT/radio-frequency integration** to maintain safety tracking even without connectivity.
False Alarms & Misuse – AI anomalies or misuse of SOS may create unnecessary alerts and hide the valid ones.	Implement **multi-layer verification with** peer confirmations, secure hold-period checks, and biometric SOS authentication.
Integration Resistance – Travel platforms may hesitate to adopt Tourist ID.	Provide lightweight APIs/SDKs and incentive models to ensure **smooth integration** without workflow disruptions.
Scalability & Performance – Handling millions of tourists may cause latency during peak seasons.	Deploy cloud-native, event-driven architecture with Kafka for streaming and Redis for caching to ensure low latency.
Data Collection for Real-Time Scoring – Collecting clean, reliable data from multiple sources can face noise or delays.	Use **AI scraper agents + LLM parsing** pipelines to structure incoming data into usable insights for the safety scoring model.

IMPACT AND BENEFITS

- Potential impact on the target audience
- Benefits of the solution (social, economic, environmental, etc.)

IMPACT AND BENEFITS

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"Tourism in India is being revolutionized by digital platforms, smart bookings, and AI-driven personalization — yet true transformation lies in ensuring safety, because a secure tourist is the foundation of a thriving tourism economy."

Tourism in India is surging—now back to **contributing around 5 % of GDP** (₹21 trillion or USD 249 billion) in 2024—and is poised to reach 10 % . Yet, safety concerns remain one of the top deterrents: nearly 27 % of potential foreign visitors cite safety and security as a major worry. Our solution bridges this critical gap with smart, real-time protection.

Impact on Target audience

- Tourists gain confidence to explore India with real-time safety scoring, multilingual alerts, and rapid response measures that protect their journey.
- Authorities & Agencies will have access to verified data streams, predictive safety scoring, and anomaly detection allows smarter deployment of resources while minimizing false positives with multi-step SOS triggering system which can't be used to trigger false alerts.
- Protecting tourists ultimately strengthens national security, as threats to visitors can quickly become threats to the nation's image and stability.
- Healthy tourism changes lives by uplifting communities, creating jobs, and fostering cultural exchange.

Benefits of the Solution

Environmental & Community Impact
Directs tourist movement intelligently, preventing overcrowding and reducing ecological strain on fragile areas while protecting local livelihoods.

Economic Impact
Safety assurance leads to higher inflows of global tourists, contributing billions economy, while creating employment opportunities.

Social Impact
AI-driven monitoring ensures solo travelers, and foreign visitors to feel equally secure while exploring India.

Strategic Impact
Real-time ensures preparedness against both natural and human-made threats.

Safety as Security
Ensuring tourist safety means safeguarding India's international image, cultural pride, and economic interests.

Tourism Transformation
India is witnessing a surge in both domestic and international tourism, driven by improved connectivity.

RESEARCH AND REFERENCES

- Details / Links of the reference and research work

RESEARCH AND REFERENCES

Developing real-time IoT-based public safety alert and emergency response systems

Enhancing Parks and Recreation Safety with Anomaly Detection

Anomaly detection in machine learning: Finding outliers for optimization of business functions

Breaking Language Barriers: NLP-Powered Emergency Communication Systems for Multilingual Communities

Safety Risk Assessment of Tourism Management System Based on PSO-BP Neural Network

Situation fencing: making geo-fencing personal and dynamic

Understanding GPS in Dynamic Geo-Fencing

Multi-Factor Risk Assessment and Route Optimization for Safe Human Travel

Unlocking the Potential of Blockchain Technology in Travel